Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following statements is false?

(A) The length of sub-tangent to the curve $x^2y^2 = 16a^4$ at the point $(-2a, 2a)$ is $2a$

(B) $x + y = 3$ is a normal to the curve $x^2 = 4y$

(C) Curves $\mathrm{y} = -4\mathrm{x}^2$ and $\mathrm{y} = \mathrm{e}^{-\mathrm{x}/2}$ are orthogonal.

(D) If $\mathrm{a} \in (-1, 0)$, then tangent at each point of the curve $y = \frac{2}{3}x^3 - 2ax^2 + 2x + 5$ makes an acute angle with the positive direction of x -axis.

Q.2

Tangent is drawn to ellipse $\frac{x^2}{27} + y^2 = 1$ at $(3\sqrt{3}\cos\theta, \sin\theta)$
[where $\theta \in (0, \pi/2)$]. Then the value of θ such that sum of intercepts on axes made by this tangent is minimum, is

(A) $\pi/3$

(B) $\pi/6$

(C) $\pi/8$

(D) $\pi/4$

Q.3

The motion of a particle is described ass $= 2 - 3\mathrm{t} + 4\mathrm{t}^3$ What is the acceleration of the particle at the point where its velocity is zero?

(A) 0

(B) 4 unit

(C) 12 unit

(D) 8 unit

Q.4

The number of tangents to the curve $x^{3/2} + y^{3/2} = 2a^{3/2}, a > 0$, which are equally inclined to the axes, is

(A) 2

(B) 1

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(C) 0

(D) 4

Q.5

The number of solutions of the equation $3\tan x + x^3 = 2$ in $\left(0, \frac{\pi}{4}\right)$. is

(A) 1

(B) 2

(C) 3

(D) infinite

Q.6

The straight line $\frac{x}{a} + \frac{y}{b} = 2$ touches the curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ at the point (a, b) for

(A) $\mathrm{n} = 1, 2$

(B) $\mathrm{n} = 3, 4, -5$

(C) $\mathrm{n} = 1, 2, 3$

(D) any value of n

Q.7

If $x\cos\theta + y\sin\theta = 2$ is perpendicular to the line $x - y = 3$, then what is one of the value of θ?

(A) $\pi/6$

(B) $\pi/4$

(C) $\pi/2$

(D) $\pi/3$

Q.8

The square of abscissa of the point on the curve $y = x^3$ where the tangent is parallel to chord joining points $1, 1$ and $5, 125$ is

(A) $\frac{1}{3}$

(B) $\frac{1}{10}$

(C) $\frac{31}{3}$

(D) $\frac{1}{5}$

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Q.9

What is the slope of the tangent to the curve $y = \sin^{-1}\left(\sin^2 x\right)$ at $x = 0$?

(A) 0

(B) 1

(C) 2

(D) None of these

Q.10

What is the slope of the tangent to the curve $x = t^2 + 3t - 8, y = 2t^2 - 2t - 5$ at $t = 2$?

(A) $7/6$

(B) $6/7$

(C) 1

(D) $5/6$

Q.11

If an equation of a tangent to the curve, $y = \cos(x + y), -1 \leq x \leq 1 + \pi$, is $x + 2y = k$ then k is equal to:

(A) 1

(B) 2

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{2}$

Q.12

The equation of the normal to the curve $y = \left|x^2 - \left|x\right|\right|$ at $x = -2.$

(A) $3y = x + 8$

(B) $x = 3y + 4$

(C) $y = 2x + 8$

(D) $y = 3x$

Q.13

If the sub-normal at any point on $y = a^{1-n}x^n$ is of constant length, then the value of n is

(A) $\frac{1}{4}$

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(B) 1

(C) $\frac{1}{3}$

(D) $\frac{1}{2}$

Q.14

Find the angle between the tangent to the curve $y^2 = 2ax$ at the points where $x = a/2$

(A) 180°

(B) 90°

(C) 0°

(D) None of these

Q.15

The total number of parallel tangents of $f_1(x) = x^2 - x + 1$ and $f_2(x) = x^3 - x^2 - 2x + 1$ is

(A) 2

(B) 0

(C) 1

(D) infinite

Q.16

How many tangents are parallel to x -axis for the curve $y = x^2 - 4x + 3$?

(A) 1

(B) 2

(C) 3

(D) No tangent is parallel to x -axis

Q.17

At what points of curve $y = \frac{2}{3}x^3 + \frac{1}{2}x^2$, the tangent makes equal angle with the x axis?

(A) $\left(\frac{1}{2}, \frac{5}{24}\right)$ and $\left(-1, -\frac{1}{6}\right)$

(B) $\left(\frac{1}{2}, \frac{4}{9}\right)$ and $(-1, 0)$

(C) $\left(\frac{1}{3}, \frac{1}{7}\right)$ and $\left(-3, \frac{1}{2}\right)$

(D) $\left(\frac{1}{3}, \frac{4}{47}\right)$ and $\left(-1, -\frac{1}{3}\right)$

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Q.18

A curve is represented by the equation $x = \sec^2 t$ and $y = \cot t$, where t is a parameter. If the tangent at the point P on the curve where $t = \pi/4$ meets the curve again at the point Q, then $|PQ|$ is equal to

(A) $\frac{5\sqrt{3}}{2}$

(B) $\frac{5\sqrt{5}}{2}$

(C) $\frac{2\sqrt{5}}{3}$

(D) $\frac{3\sqrt{5}}{2}$

Q.19

Area of the triangle formed by the normal to the curve $x = e^{\sin y}$ at $(1, 0)$ with the coordinate axes is:

(A) $1/4$

(B) $1/2$

(C) $3/4$

(D) 1

Q.20

The two curves $x^3 - 3xy^2 + 2 = 0$ and $3x^2y - y^3 = 2$

(A) cuts at right angle

(B) touch each other

(C) cut at an angle $\frac{\pi}{3}$

(D) cut at an angle $\frac{\pi}{4}$

Answers & Solutions

Q.1 Answer:

Curves $\mathrm{y} = -4\mathrm{x}^2$ and $\mathrm{y} = \mathrm{e}^{-\mathrm{x}/2}$ are orthogonal.

Solution:

(a) $x^2y^2 = 16a^4$

$L_{ST} = \left|\frac{y}{x_T}\right| \Rightarrow xy = 4a^2$

$y + xy' = 0$

$y' = \frac{-y}{x}; L_{ST} = \left|\frac{y}{y/x}\right| = x \Rightarrow L_{ST} = 2a$

$\therefore$ **(a) is true.**

(b) $\frac{dy}{dx} = \frac{x}{2} = 1$ **i.e.** $x = 2$

$\therefore (2, 1)$ **is the point on the curve** $\mathrm{x}^2 = 4\mathrm{y}$ **at which the normal is** $y - 1 = -1(x - 2)$ **i.e.** $x +$

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$y = 3 \quad \therefore$ **(b) is true**

(c) $y = -4x^2, y = e^{-x/2}$

The curves are non-intersecting

$\therefore$ curves are not orthogonal i.e. (c) is false.

(d) $y = \frac{2}{3}x^3 - 2ax^2 + 2x + 5$

$\frac{dy}{dx} = 2x^2 - 4ax + 2 = 2\left(x^2 - 2ax + 1\right)$

$= 2(x-a)^2 + 2 - 2a^2 > 0$

$\left[\because a \in (-1, 0) \Rightarrow 0 < a^2 < 1\right]$

$\therefore$ (d) is true

Q.2 Answer:

$\pi/6$

Solution:

$\frac{x\cos\theta}{3\sqrt{3}} + y\sin\theta = 1$

Sum of intercepts $3\sqrt{3}\sec\theta + \operatorname{cosec}\theta = f(\theta)$ **(say)**

$f'(\theta) = \frac{3\sqrt{3}\sin^3\theta - \cos^3\theta}{\sin^2\theta\cos^2\theta}$

$\Rightarrow At\theta = \pi/6, f(\theta)$ **is minimum.**

Q.3 Answer:

12 unit

Solution:

Given rule is:

Distance, $s = 2 - 3t + 4t^3$

$\Rightarrow$ **Velocity** $= \frac{\mathrm{ds}}{\mathrm{dt}} = -3 + 12\mathrm{t}^2$

$\Rightarrow$ **Acceleration** $= \frac{\mathrm{d}^2\mathrm{s}}{\mathrm{dt}^2} = 24\mathrm{t}$

Since, velocity is zero $\frac{\mathrm{ds}}{\mathrm{dt}} = 0; \Rightarrow 0 = -3 + 12\mathrm{t}^2$

$\Rightarrow \mathrm{t} = \frac{1}{2}$

Acceleration (when velocity is zero) $= 24 \times \frac{1}{2} = 12$

Q.4 Answer:

1

Solution:

Given curve is $x^{3/2} + y^{3/2} = 2a^{3/2} \quad \ldots$**(1)**

$\therefore \frac{3}{2}\sqrt{x} + \frac{3}{2}\sqrt{y}\frac{dy}{dx} = 0$ **or** $\frac{dy}{dx} = -\frac{\sqrt{x}}{\sqrt{y}}$

Since the tangent is equally inclined to the axes, $\frac{dy}{dx} = \pm 1 \quad \therefore -\frac{\sqrt{x}}{\sqrt{y}} = \pm 1$ **or** $-\frac{\sqrt{x}}{\sqrt{y}} = -1$

$\therefore \sqrt{x} = \sqrt{y} \quad [\because \sqrt{x} > 0, \sqrt{y} > 0]$

Putting $\sqrt{y} = \sqrt{x}$ **in** (1)**, we get** $2x^{3/2} = 2a^{3/2}$ **or** $x^3 = a^3$

Therefore, $x = a$ **and, so,** $y = a$**.**

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Q.5 Answer:

1

Solution:

Let $f(x) = 3\tan x + x^3 - 2$
Then $f'(x) = 3\sec^2 x + 3x^2 > 0$. Hence, $f(x)$
increases. Also, $f(0) = -2$ and $f\left(\frac{\pi}{4}\right) > 0$.
So, by intermediate value theorem, $f(c) = 2$ for some $c \in \left(0, \frac{\pi}{4}\right)$.
Hence, $f(x) = 0$ has only one root.

Q.6 Answer:

any value of n

Solution:

The point (a, b) lies on both the straight line and the given curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$.
Differentiating the equation, we get $\frac{dy}{dx} = -\frac{x^{n-1}}{a^n} \cdot \frac{b^n}{y^{n-1}}$
$\therefore \left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)_{\text{at (a,b)}} = -\frac{\mathrm{b}}{\mathrm{a}}$ **= the slope of $\frac{\mathrm{x}}{\mathrm{a}} + \frac{\mathrm{y}}{\mathrm{b}} = 2$**
Hence, it touches the curve at (a, b) whatever may be the value of n.

Q.7 Answer:

$\pi/4$

Solution:

Consider a line $x\cos\theta + y\sin\theta = 2$
$\Rightarrow \quad y\sin\theta = -x\cos\theta + 2$
$\Rightarrow \quad y = -x\frac{\cos\theta}{\sin\theta} + \frac{2}{\sin\theta}$
$\Rightarrow \quad y = -x\cot\theta + 2\,\mathrm{cosec}\,\theta$
On comparing this equation with $y = mx + c$ we get slope of line $x\cos\theta + y\sin\theta = 2$ is
$-\cot\theta$
Also, we have a line $x - y = 3 \Rightarrow \quad y = x - 3$
slope of line $x - y = 3$ is 1 . Since, both the lines are perpendicular to each other
$\therefore$ Product of their slopes $= -1$
$\Rightarrow \quad (-\cot\theta)(1) = -1$
$\Rightarrow \quad \cot\theta = 1 = \cot\frac{\pi}{4}$
$\Rightarrow \quad \theta = \frac{\pi}{4}$

Q.8 Answer:

$\frac{31}{3}$

Solution:

We have,
$y = x^3$

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$\Rightarrow \frac{dy}{dx} = 3x^2$

Let the point of contact of the tangent to the given curve be α, β then slope of tangent is

$m = \frac{dy}{dx}_{\alpha, \beta} = 3\alpha^2$

Slope of chord joining $1, 1$ and $5, 125$ is

$\frac{125 - 1}{5 - 1} = 31$

Therefore, slope of chord is 31.

For tangent to be parallel to chord

$m = 31$

$\Rightarrow 3\alpha^2 = 31$

$\Rightarrow \alpha^2 = \frac{31}{3}$

Q.9 Answer:

0

Solution:

$y = \sin^{-1}\left(\sin^2 x\right)$

$\frac{dy}{dx} = \frac{2 \sin x \cos x}{\sqrt{1 - \sin^4 x}} \Rightarrow \frac{dy}{dx} = \frac{\sin 2x}{\sqrt{1 - \sin^4 x}}$

at $\mathrm{x} = 0, \frac{\mathrm{dy}}{\mathrm{dx}} = 0$

Q.10 Answer:

$6/7$

Solution:

$\frac{\mathrm{dx}}{\mathrm{dt}} = 2\mathrm{t} + 3, \frac{\mathrm{dy}}{\mathrm{dt}} = 4\mathrm{t} - 2$

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{4t-2}{2t+3}$

slope at $\mathrm{t} = 2 = \left.\frac{\mathrm{dy}}{\mathrm{dx}}\right|_{\mathrm{t}=2} = \frac{4(2)-2}{2(2)+3} = \frac{6}{7}$

Q.11 Answer:

$\frac{\pi}{2}$

Solution:

Let $y = \cos(x + y)$

$\Rightarrow \frac{dy}{dx} = -\sin(x + y)\left(1 + \frac{dy}{dx}\right)$ **...(1)**

Now, given equation of tangent is $x + 2y = k \Rightarrow$ **Slope** $= \frac{-1}{2}$ **So,** $\frac{dy}{dx} = \frac{-1}{2}$ **put this value in** (1)**, we get**

$\frac{-1}{2} = -\sin(x + y)\left(1 - \frac{1}{2}\right) \Rightarrow \sin(x + y) = 1$

$\Rightarrow x + y = \frac{\pi}{2} \Rightarrow y = \frac{\pi}{2} - x$

Now, $\frac{\pi}{2} - x = \cos(x + y) \Rightarrow x = \frac{\pi}{2}$ **and** $y = 0$

Thus $x + 2y = k \Rightarrow \frac{\pi}{2} = k$

Q.12 Answer:

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$3y = x + 8$

Solution:

In the neighbourhood of $x = -2, y = x^2 + x.$

Hence, the point on curve is $(-2, 2)$

$\frac{dy}{dx} = 2x + 1$ **or** $\left.\frac{dy}{dx}\right|_{x=-2} = -3$

So, the slope of the normal at $(-2, 2)$ **is** $\frac{1}{3}$.

Hence, the equation of the normal is $\frac{1}{3}(x + 2) = y - 2$ **or** $3y = x + 8$

Q.13 Answer:

$\frac{1}{2}$

Solution:

$y = a^{1-n}x^n$ **or** $\frac{dy}{dx} = a^{1-n}nx^{n-1}$

Sub-normal $= \left|y\frac{dy}{dx}\right| = \left|ya^{1-n}nx^{n-1}\right|$

$= \left|a^{1-n}x^n a^{1-n}nx^{n-1}\right| = \left|a^{2-2n}x^{2n-1}\right|$

which is constant if $2n - 1 = 0$ **or** $n = 1/2.$

Q.14 Answer:

90°

Solution:

We have, $y^2 = 2ax$ **...(i)**

$\text{Put } x = \frac{a}{2}; y^2 = 2a\left(\frac{a}{2}\right) \Rightarrow y = \pm a$

$\therefore$ **The points are** $\left(\frac{\mathrm{a}}{2}, \mathrm{a}\right)$ **and** $\left(\frac{\mathrm{a}}{2}, -\mathrm{a}\right)$

Differentiating (i) with respect to x, we get

$2y\frac{dy}{dx} = 2a \Rightarrow \frac{dy}{dx} = \frac{a}{y}$

(t $\left(\frac{a}{2}, a\right)$; $\frac{dy}{dx} = \frac{a}{y} = \frac{a}{a} = 1 = m_1$ **(say)**

At $\left(\frac{a}{2}, -a\right)$; $\frac{dy}{dx} = \frac{a}{y} = \frac{a}{-a} = -1 = m_2$ **(say)**

Since $\mathrm{m}_1\ \mathrm{m}_2 = -1$**, the two tangents are at right angles.**

Q.15 Answer:

infinite

Solution:

Here,

$f_1(x) = x^2 - x + 1$ **and** $f_2(x) = x^3 - x^2 - 2x + 1$

or $f_1'(x_1) = 2x_1 - 1$ **and** $f_2'(x_2) = 3x_2^2 - 2x_2 - 2$

Let the tangents drawn to the curves $y = f_1(x)$ **and** $y = f_2(x)$ **at** $(x_1, f_1(x_1))$ **and** $(x_2, f_2(x_2))$ **be parallel. Then**

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$2x_1 - 1 = 3x_2^2 - 2x_2 - 2$ or $2x_1 = \left(3x_2^2 - 2x_2 - 1\right)$

So, which is possible for infinite numbers of ordered pairs. So, there are infinite solutions.

Q.16 Answer:

1

Solution:

Let $y = x^2 - 4x + 3$

Differentiate both sides w.r.t. ' x **'**

$\frac{dy}{dx} = 2x - 4$

So, slope $= 2x - 3$ **Since, tangent is** $\|$ **to** x **-axis**

$\therefore$ **slope** $= 0$

$\Rightarrow \frac{\mathrm{dy}}{\mathrm{dx}} = 0 \Rightarrow 2\mathrm{x} - 4 = 0 \Rightarrow \mathrm{x} = \frac{4}{2}$

$\Rightarrow$ **one tangent**

Q.17 Answer:

$\left(\frac{1}{2}, \frac{5}{24}\right)$ and $\left(-1, -\frac{1}{6}\right)$

Solution:

Given equation of curve is

$$y = \frac{2}{3}x^3 + \frac{1}{2}x^2$$

$$\frac{dy}{dx} = 2x^2 + x$$

Since, the tangent makes equal angle with x-axis,

$$\frac{\mathrm{dy}}{\mathrm{dx}} = 1$$

$$\Rightarrow 2\mathrm{x}^2 + \mathrm{x} - 1 = 0$$

$$\Rightarrow \mathrm{x} = -1, \frac{1}{2}$$

At $x = -1 \Rightarrow y = -\frac{1}{6}$

At $x = \frac{1}{2} \Rightarrow y = \frac{5}{24}$

Q.18 Answer:

$\frac{3\sqrt{5}}{2}$

Solution:

$$x = \sec^2 t$$

$$y = \cot t$$

$$\frac{dy}{dx} = -\frac{\csc^2 t}{2\sec^2 t \cdot \tan t} = -\frac{\cot^3 t}{2}$$

Now,

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$$\left.\frac{dy}{dx}\right|_{t=45^\circ} = -\frac{1}{2}$$

$$x|_{t=45^\circ} = 2$$

$$y|_{t=45^\circ} = 1$$

Hence the equation of the tangent will be

$y - 1 = -\frac{1}{2}(x-2) \Rightarrow 2y - 2 = -x + 2 \Rightarrow x + 2y = 4 \ldots$ (i)

Now,

$$\sec^2 t = 1 + \tan^2 t$$

$$\sec^2 t = 1 + \frac{1}{\cot^2 t}$$

Or

$x = 1 + \frac{1}{y^2} \Rightarrow xy^2 = y^2 + 1$ **is the equation of curve.**

Solving the equation of tangent and equation of curve we get $4 - 2y = 1 + \frac{1}{y^2}$

Or

$4y^2 - 2y^3 = y^2 + 1 \Rightarrow 2y^3 - 3y^2 + 1 = 0$

$y = -\frac{1}{2}$ **and** $y = 1$

For $y = 1; x = 2$

For $y = -\frac{1}{2}; x = 5$

Hence,

$$Q = \left(5, -\frac{1}{2}\right) \text{ and } P = (2, 1)$$

$$\therefore PQ = \frac{3\sqrt{5}}{2}.$$

Q.19 Answer:

$1/2$

Solution:

$x = e^{\sin y} \Rightarrow \ln x = \sin y \Rightarrow \frac{1}{x}\frac{dx}{dy} = \cos y$

$\therefore$ **slope of normal at** $(1, 0)$

$= -\frac{dx}{dy} = -1 \times \cos 0 = -1$

$\therefore$ **Equation of normal is** $y - 0 = -1(x - 1)$

$\Rightarrow x + y = 1$

Which forms triangle of area $\frac{1}{2}$ **with axes.**

Q.20 Answer:

cuts at right angle

Solution:

Two curves cuts at right angle if product of their slopes is -1.

Two given curves are

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$x^3 - 3xy^2 + 2 = 0$ ……(1)

and $3x^2y - y^3 - 2 = 0$ ……(2)

Differentiate equ. (i),

$3x^2 - 3\left[y^2 + 2xy\frac{dy}{dx}\right] = 0$

$\Rightarrow 3\left(x^2 - y^2\right) = 6xy\frac{dy}{dx}$

$\Rightarrow m_1 = \frac{dy}{dx} = \frac{3\left(x^2-y^2\right)}{6xy}$

Differcntiate equ. (ii),

$3x^2y - y^3 - 2 = 0$

$\Rightarrow 3\left[x^2\frac{dy}{dx} + 2xy\right] - 3y^2\frac{dy}{dx} = 0$

$\Rightarrow x^2\frac{dy}{dx} + 2xy - y^2\frac{dy}{dx} = 0$

$\Rightarrow \left(x^2 - y^2\right)\frac{dy}{dx} = -2xy$

$\Rightarrow m_2 = \frac{dy}{dx} = \frac{-2xy}{(x^2-y^2)}$

$m_1 \times m_2 = \frac{\left(x^2-y^2\right)}{2xy} \times \frac{-2xy}{(x^2-y^2)}$

$\Rightarrow m_1 \times m_2 = -1$

i.e. curves cuts at right angle.

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